real difference among the various kinds of light is the frequency of the waves in each band. In fact, credit Hertz for recognizing that there is such a thing as an electromagnetic spectrum. In his honor, the unit of frequency—in waves per second—for anything that vibrates, including sound, has duly been named the hertz.

Mysteriously, astrophysicists were a bit slow to make the connection between the newfound invisible bands of light and the idea of building a telescope that might see those bands from cosmic sources. Delays in detector technology surely mattered here. But hubris must take some of the blame: how could the universe possibly send us light that our marvelous eyes cannot see? For more than three centuries—from Galileo's day until Edwin Hubble's—building a telescope meant only one thing: making an instrument to catch visible light, enhancing our biologically endowed vision.

A telescope is merely a tool to augment our meager senses, enabling us to get better acquainted with faraway places. The bigger the telescope, the dimmer the objects it brings into view; the more perfectly shaped its mirrors, the sharper the image it makes; the more sensitive its detectors, the more efficient its observations. But in all cases, every bit of information a telescope delivers to the astrophysicist comes to Earth on a beam of light.

Celestial happenings, however, don't limit themselves to what's convenient for the human retina. Instead, they typically emit varying amounts of light simultaneously in multiple bands. So without telescopes and their detectors tuned across the entire spectrum, astrophysicists would remain blissfully ignorant of some mind-blowing stuff in the universe.

Take an exploding star—a supernova. It's a cosmically common and seriously high-energy event that generates prodigious quantities of X-rays. Sometimes, bursts of gamma rays and flashes of ultraviolet accompany the explosions, and there's never a shortage of visible light. Long after the explosive gases cool, the shock waves dissipate, and the visible light fades, the supernova "remnant" keeps on shining in the infrared, while pulsing in radio waves. That's where pulsars come from, the most reliable timekeepers in the universe.

Most stellar explosions take place in distant galaxies, but if a star were to blow up within the Milky Way, its death throes would be bright enough for everyone to see, even without a telescope. But nobody on Earth saw the invisible X-rays or gamma rays from the last two